Chapter 3

3.1 P-N Junction

A p-n junction is a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

The p-side or the positive side of the semiconductor has an excess of holes and the n-side or the negative side has an excess of electrons. In a semiconductor, the p-n junction is created by the method of doping. The process of doping is explained in further detail in the next section.

3.1.1 Formation of P-N Junction

If different semiconductor materials are used to make a p-n junction, there will be a grain boundary that would prevent the movement of electrons from one side to the other by scattering the electrons and holes and thus, we use the process of doping.Consider a thin p-type silicon semiconductor sheet. If we add a small amount of pentavalent impurity to this, a part of the p-type Si (Silicon) will get converted to n-type silicon. This sheet will now contain both p-type region and n-type region and a junction between these two regions. The processes that follow after the formation of a p-n junction are of two types – diffusion and drift. As we know, there is a difference in the concentration of holes and electrons at the two sides of a junction, the holes from the p-side diffuse to the n-side and the electrons from the n-side diffuse to the p-side. These give rise to a diffusion current across the junction.

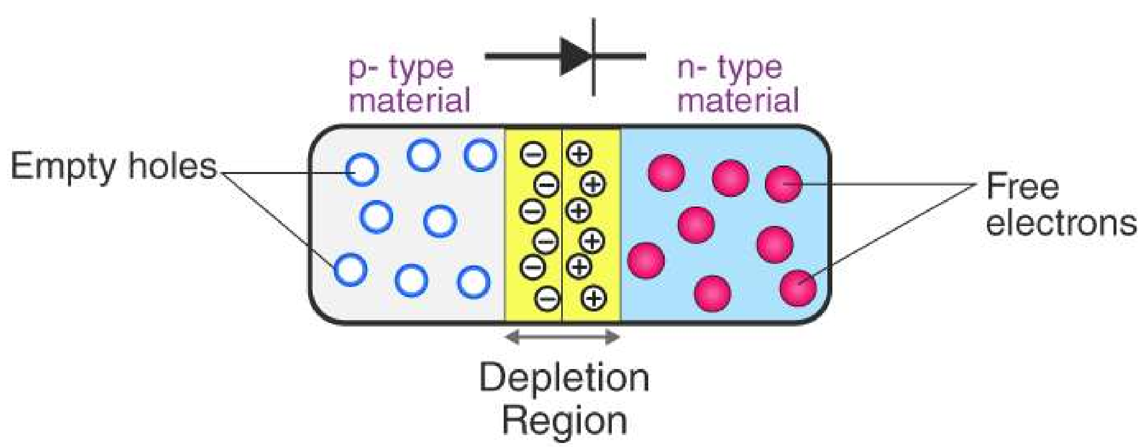


Fig. 3.1 PN junction

When an electron diffuses from the n-side to the p-side, an ionized donor is left behind on the n-side, which is immovable. As the process goes on, a layer of positive charge is developed on the n-side of the junction. Similarly, when a hole goes from the p-side to the n-side, and ionized acceptor is left behind in the p-side, resulting in the formation of a layer of negative charges in the p-side of the junction. This region of positive charge and negative charge on either side of the junction is termed as the depletion region. Due to this positive space charge region on either side of the junction, an electric field direction from a positive charge towards the negative charge is developed. Due to this electric field, an electron on the p-side of the junction moves to the n-side of the junction. This motion is termed as the drift. The direction of drift current is opposite to that of the diffusion current.

3.2 Diode

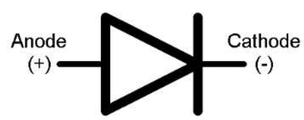
A diode is defined as a two-terminal electronic component that only conducts current in one direction (as long as it is operated within a specified voltage level). An ideal diode will have zero resistance in one direction, and infinite resistance in the reverse direction.
Although in the real world, diodes cannot achieve zero or infinite resistance. Instead, a diode will have negligible resistance in one direction (to allow current flow), and very high resistance in the reverse direction (to prevent current flow).
Semiconductor diodes are the most common type of diode. These diodes begin conducting electricity only if a certain threshold voltage is present in the forward direction (i.e., the “low resistance” direction). The diode is said to be “forward biased” when conducting current in this direction. When connected within a circuit in the reverse direction (i.e., the “high resistance” direction), the diode is said to be “reverse biased”.
A diode only blocks current in the reverse direction (i.e., when it is reverse biased) while the reverse voltage is within a specified range. Above this range, the reverse barrier breaks. The voltage at which this breakdown occurs is called the “reverse breakdown voltage”.

When the voltage of the circuit is higher than the reverse breakdown voltage, the diode is able to conduct electricity in the reverse direction (i.e., the "high resistance" direction). This is why we say diodes have a high resistance in the reverse direction – not an infinite resistance.

A P-N junction is the simplest form of the semiconductor diode. In ideal conditions, this P-N junction behaves as a short circuit when it is forward biased, and as an open circuit when it is in the reverse biased.

Diode Symbol

The symbol of a diode is shown below. The arrowhead points in the direction of conventional current flow in the forward biased condition. That means the anode is connected to the p side and the cathode is connected to the n side.



Symbol of Diode

We can create a simple P-N junction diode by doping pentavalent or donor impurity in one portion and trivalent or acceptor impurity in the other portion of silicon or germanium crystal block.

These dopings make a PN junction in the middle part of the block. We can also form a P-N junction by joining a p-type semiconductor and n-type semiconductor together with a special fabrication technique. The terminal connected to the p-type is the anode. The terminal connected to the n-type side is the cathode.

3.2.1 Working Principle of a Diode

A diode's working principle depends on the interaction of n-type and p-type semiconductors. An n-type semiconductor has plenty of free electrons and a very few numbers of holes. In other words, we can say that the concentration of free electrons is high and that of holes is very low in an n-type semiconductor. Free electrons in the n-type semiconductor are referred to as majority charge carriers, and holes in the n-type semiconductor are referred to as minority charge carriers.

A p-type semiconductor has a high concentration of holes and a low concentration of free electrons. Holes in the p-type semiconductor are majority charge carriers, and free electrons in the p-type semiconductor are minority charge carriers.

3.2.2 Forward Biased Diode

A diode is forward biased if a positive terminal of a source is connected to the p-type side and the negative terminal of the source is connected to the n-type side of the diode. Initially, there is no current flowing through the diodebecause the majority charge carriers still do not get sufficient impact of the external field to cross the depletion region. As we know that the depletion region acts as a potential barrier against the majority charge carriers. This potential barrier is called forward potential barrier. The majority charge carriers start crossing the forward potential barrier only when the value of externally applied voltage across the junction is more than the potential of the forward barrier. For silicon diodes, the forward barrier potential is 0.7 volt and for germanium diodes, it is 0.3 volt.

When the externally applied forward voltage across the diode becomes more than the forward barrier potential, the free majority charge carriers start crossing the barrier and contribute the forward diode current. In this situation, the diode behaves as a short-circuited path, and the forward current gets limited by only externally connected resistors to the diode.

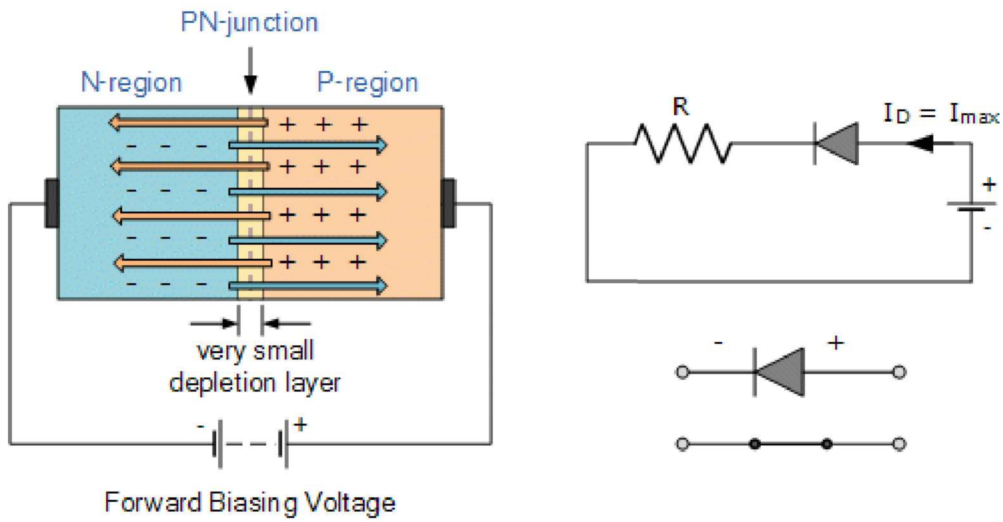


Fig. 3.2 Forward biased diode

3.2.3 Reverse Biased Diode

A diode is reverse biased ifthe negative terminal of the voltage source is connected to the p-type side and the positive terminal of the voltage source to the n-type side of the diode. Due to the electrostatic attraction of the negative potential of the source, the holes in the p-type region would be shifted more away from the junction leaving more uncovered negative ions at the junction. In the same way, the free electrons in the n-type region would be shifted more away from the junction towards the positive terminal of the voltage source leaving more uncovered positive ions in the junction.

As a result of this phenomenon, the depletion region becomes wider. This condition of a diode is called the reverse biased condition. At that condition, no majority carriers cross the junction, and they instead move away from the junction. In this way, a diode blocks the flow of current when it is reverse biased.

As we know that there are always some free electrons in the p-type semiconductor and some holes in the n-type semiconductor. These opposite charge carriers in a semiconductor are called minority charge carriers.In the reverse biased condition, the holes find themselves in the n-type side would easily cross the reverse-biased depletion region as the field across the depletion region does not present rather it helps minority charge carriers to cross the depletion region. As a result, there is a tiny current flowing through the diode from positive to the negative side. The amplitude of this current is very small as the number of minority charge carriers in the diode is very small. This current is called reverse saturation current.

If the reverse voltage across a diode is increased beyond a safe value, a number of covalent bonds are broken to contribute a huge number of free electron-hole pairs in the diode. This is due to higher electrostatic force and due to higher kinetic energy of minority charge carriers colliding with atoms. The huge number of such generated charge carriers would contribute a huge reverse current in the diode. If this current is not limited by an external resistance connected to the diode circuit, the diode may permanently be destroyed. This phenomenon is called *Avalanche Breakdown.*

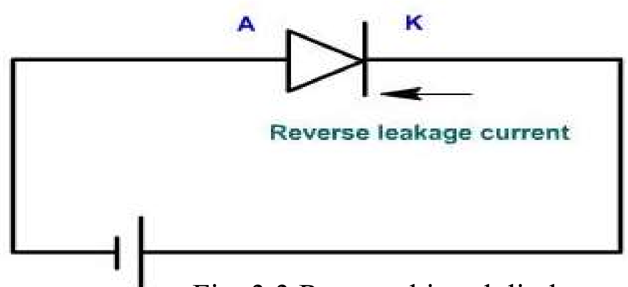


Fig. 3.3 Reverse biased diode

3.2.4 IV **Characteristics of a Diode**

Curve in Figure 3.3 indicates Current vs Voltage relationship in a diode. The forward portion of the curve indicates that the diode conducts when the P-region is made positive and the N-region negative.The diode conducts almost no current in the high resistance direction, i.e., when the P region is made negative and the N-region is made positive. Now the holes and electrons are drained away from the junction, causing the barrier potential to increase. This condition is indicated by the reverse current portion of the curve.The dotted section of the curve indicates the ideal curve, which would result if it were not for avalanche breakdown. Figure 3.3 shows the static characteristic of a junction diode.

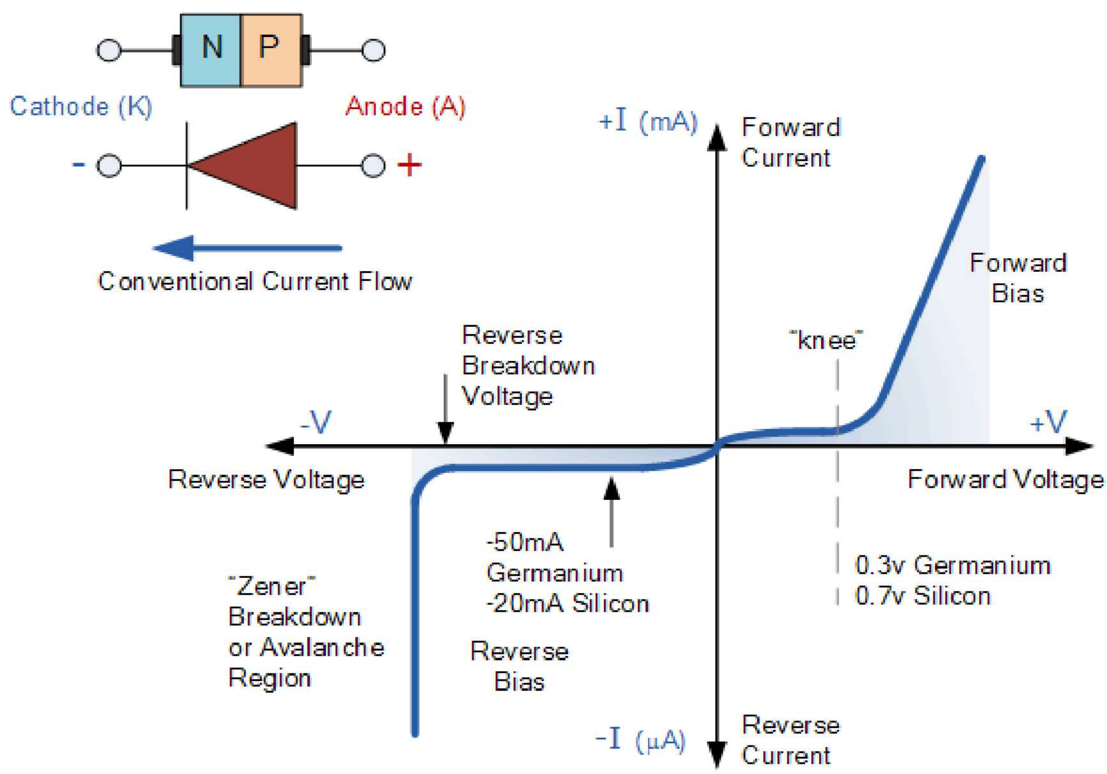


Fig. 3.3 IV characteristics of a diode

3.2.5 Rectification Circuits

The main application of p-n junction diode is in rectification circuits. These circuits are used to convert AC signals to DC signals. Diode rectifier gives an alternating voltage which pulsates in accordance with time.

There are two primary methods of diode rectification:

- Half Wave Rectifier
- Full Wave Rectifier

Half Wave Rectifier

In a half-wave rectifier, one half of each AC input cycle is rectified. When the p-n junction diode is forward biased, it offers a little resistance and when it is reversing biased it provides high resistance. During one-half cycles, the diode is forward biased when the input voltage is applied and in the opposite half cycle, it is reverse biased.

The half-wave rectifier has both positive and negative cycles. During the positive half of the input, the current will flow from positive to negative which will generate only a positive half cycle of the AC supply. When AC supply is applied to the transformer, the voltage will be decreasing at the secondary winding of the diode. All the variations in the AC supply will reduce, and we will get the pulsating DC voltage to the load resistor.
In the second half cycle, the current will flow from negative to positive and the diode will be reverse biased. Thus, at the output side, there will be no current generated, and we cannot get power at the load resistance. A small amount of reverse current will flow during reverse bias due to minority carriers.

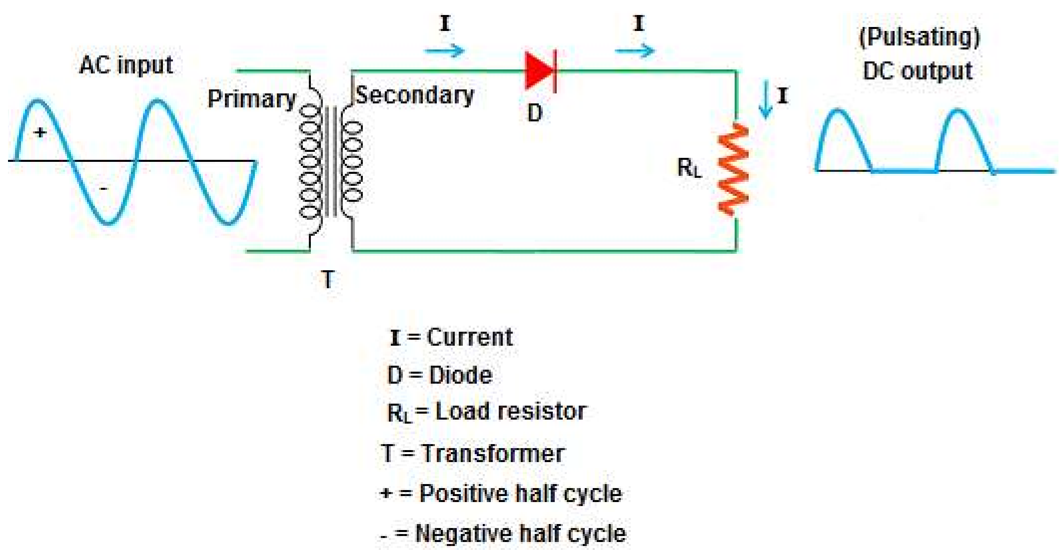


Fig. 3.4 Half wave rectifier

Full Wave Rectifier

Full-wave rectifier circuits are used for producing an output voltage or output current which is purely DC. The main advantage of a full-wave rectifier over half-wave rectifier is that the average output voltage is higher in full-wave rectifier.Ripple produced in full-wave rectifier is less as compared to the half-wave rectifier.
The full-wave rectifier utilizes both halves of each AC input. When the p-n junction is forward biased, the diode offers low resistance and when it is reversing biased it gives

high resistance. The circuit is designed in such a manner that in the first half cycle if the diode is forward biased then in the second half cycle it is reverse biased and so on.

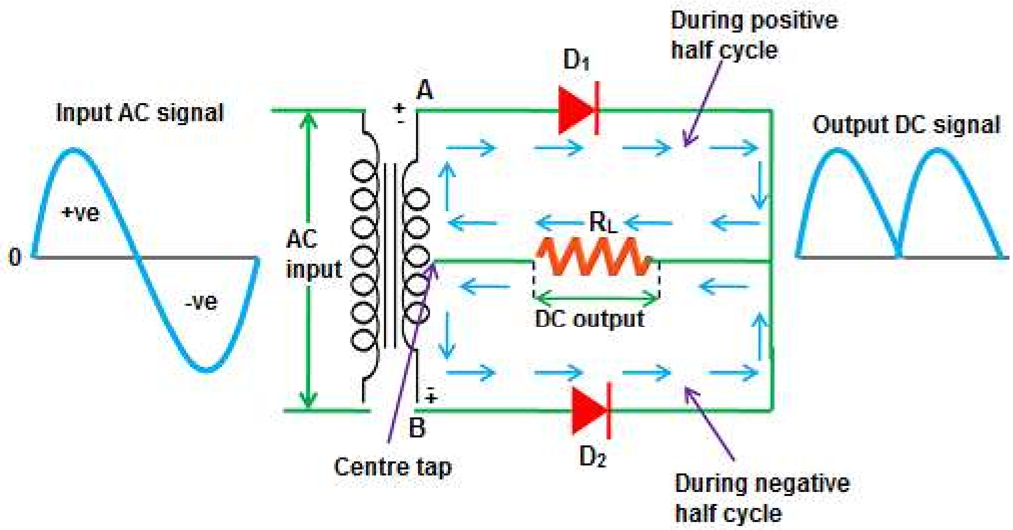


Fig. 3.5 Full wave rectifier

Do you know?

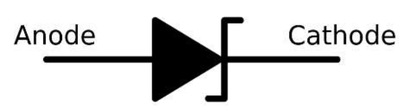
Filtering capacitors are used to convert pulsating DC voltage into a constant DC voltage.

Point to Ponder

- Where are rectifiers being used?

3.3 Zener Diode

A heavily doped semiconductor diode which is designed to operate in reverse direction is known as the Zener diode. In other words, the diode which is specially designed for optimizing the breakdown region is known as the Zener diode. The symbolic representation of Zener diode is shown in the figure.

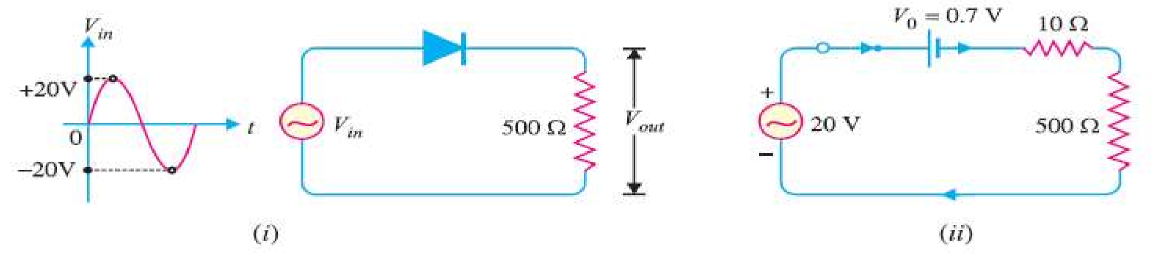


Symbol of a Zener Diode

Example 3.1

An AC voltage of peak value 20 V is connected in series with a silicon diode and load resistance of 500 Ω. If the forward resistance of diode is 10 Ω, Find:

- (i) peak current through diode
- (ii) peak output voltage
- (iii) What will be these values if the diode is assumed to be ideal?



Solution:

Peak input voltage = 20 V

Forward resistance, r_f= 10 Ω

Load resistance, R_L= 500 Ω

Potential barrier voltage, V_0 = 0.7 V

The diode will conduct during the positive half-cycles of AC input voltage only.

V_4 = 0.2 X 20 = 4V

- (i) The peak current through the diode will occur at the instant when the input voltage reaches positive peak i.e. $V_{in} = V_F = 20$ V.

$$\therefore \quad V_F = V_0 + (I_f)_{peak}\,[r_f + R_L] \qquad \ldots(i)$$

$$\text{or} \quad (I_f)_{peak} = \frac{V_F - V_0}{r_f + R_L} = \frac{20 - 0.7}{10 + 500} = \frac{19.3}{510}\text{ A} = \mathbf{37.8\ mA}$$

- (ii) Peak output voltage:

$$\text{Peak output voltage} = (I_f)_{peak} \times R_L = 37.8\text{ mA} \times 500\ \Omega = \mathbf{18.9\ V}$$

- (iii) Ideal diode case:

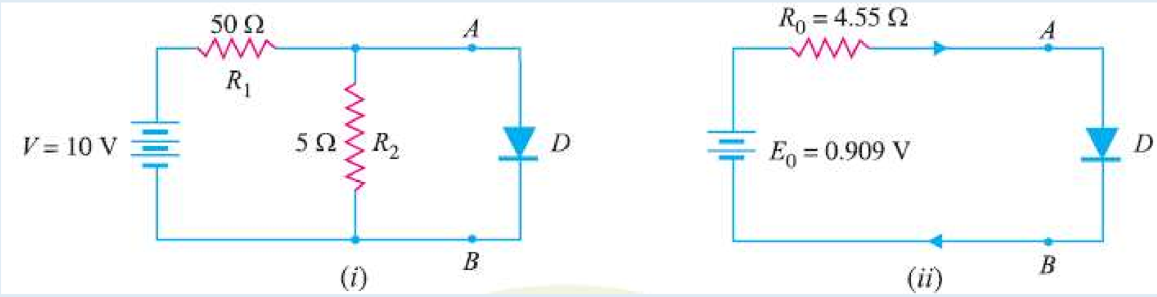
$$\text{or} \quad (I_f)_{peak} = \frac{V_F}{R_L} = \frac{20\text{ V}}{500\ \Omega} = \mathbf{40\ mA}$$

$$\text{Peak output voltage} = (I_f)_{peak} \times R_L = 40\text{ mA} \times 500\ \Omega = \mathbf{20\ V}$$

Example 3.2

Find the current through the diode in the circuit shown:

(i) Assume the diode to be ideal.



Solution:

We shall use Thevenin's theorem to find current in the diode.

E_0 = Thevenin's voltage

= Open circuited voltage across AB with diode removed

$$= \frac{R_2}{R_1 + R_2} \times V = \frac{5}{50+5} \times 10 = 0.909\text{ V}$$

R_0 = Thevenin's resistance

= Resistance at terminals AB with diode removed and battery replaced by a short circuit

$$= \frac{R_1 R_2}{R_1 + R_2} = \frac{50 \times 5}{50+5} = 4.55\ \Omega$$

Fig shows Thevenin's equivalent circuit. Since the diode is ideal, it has zero resistance

$$\therefore \quad \text{Current through diode} = \frac{E_0}{R_0} = \frac{0.909}{4.55} = 0.2\text{ A} = \mathbf{200\ mA}$$

3.3.1 Zener Diode Circuit Diagram:

The circuit diagram of the Zener diode is shown in the figure below. The Zener diode is used in reverse bias. The reverse biasing means the n-type material of the diode is connected to the positive terminal of the supply and the P-type material is connected to the negative terminal of the supply. The depletion region of the diode is very thin because it is made of the heavily doped semiconductor material.

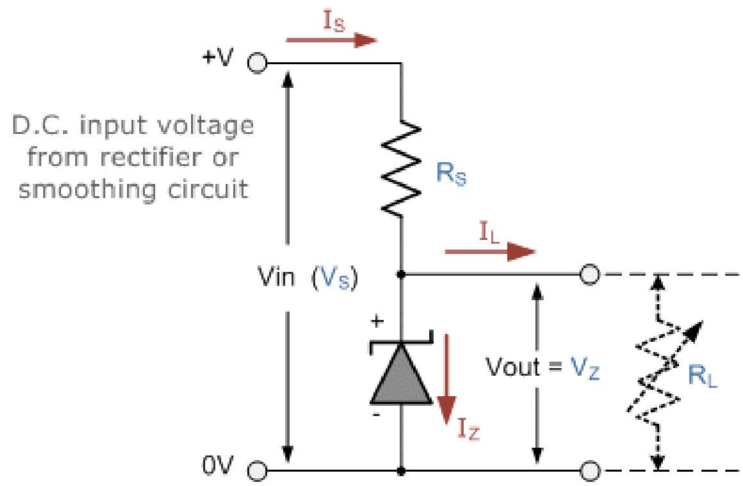


Fig. 3.6 Zener diode circuit diagram

3.3.2 **Working of Zener Diode**

The Zener diode is made up of heavily doped semiconductor material. The heavily doped means the high-level impurities is added to the material for making it more conductive. The depletion region of the Zener diode is very thin because of the impurities. The heavily doping material increases the intensity of the electric field across the depletion region of the Zener diode even for the small reverse voltage.

When no biasing is applied across the Zener diode, the electrons remain in the valence band of the p-type material and no current flow through the diode. The band in which the valence electrons (outermost orbit electron) place is known as the valence band electron. The electrons of the valence band easily move from one band to another when the external energy is applied across it.

When the reverse bias applies across the diode and the supply voltage is equal to the Zener voltage then it starts conducting in the reverse bias direction. The Zener voltage is the voltage at which the depletion region completely vanishes.

The reverse bias applies across the diode increases the intensity of electric field across the depletion region. Thus, it allows the electrons to move from the valence band of P-type material to the conduction band of N-type material. This transferring of valence band electrons to the conduction band reduces the barrier between the p and n-type material. When the depletion region become completely vanish the diode starts conducting in the reverse biased.

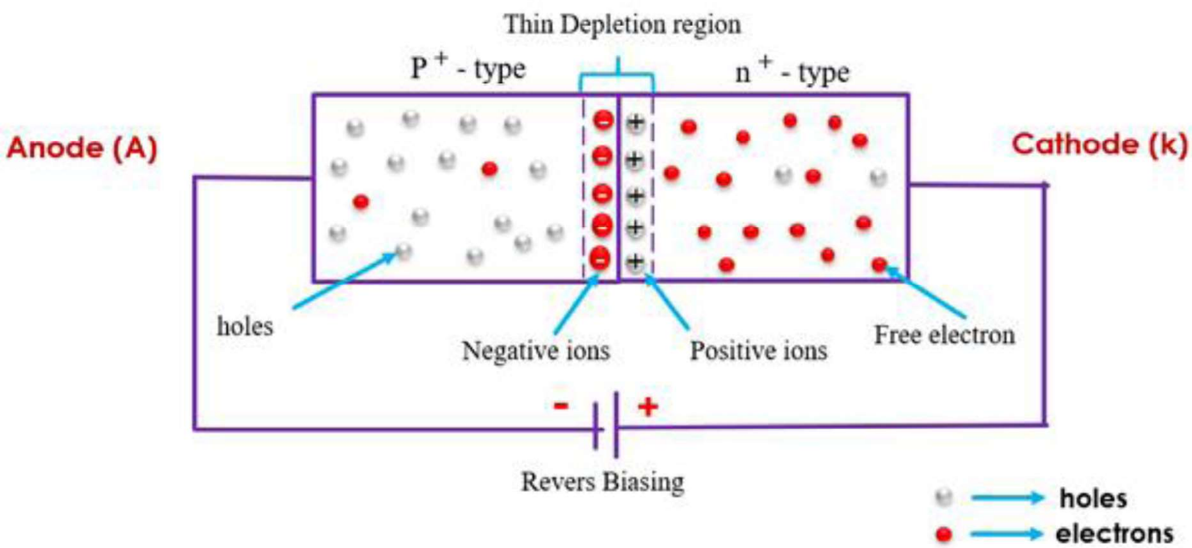


Fig. 3.7 Working of a Zener diode

3.3.3 Characteristics of Zener Diode

The VI characteristic graph of the Zener diode is shown in the figure below. This curve shows that the Zener diode, when connected in forwarding bias, behaves like an ordinary diode. But when the reverse voltage applies across it and the reverse voltage rises beyond the predetermined rating, the Zener breakdown occurs in the diode.

At Zener breakdown voltage the current starts flowing in the reverse direction. The graph of the Zener breakdown is not exactly vertical shown above which shows that the Zener diode has resistance. The voltage across the Zener is represented by the equation shown below.

$$V = V_Z + I_Z R_Z$$

Interesting Information

- Zener is named after American physicist Clarence Zener, who first described the Zener effect in 1934 in his primarily theoretical studies of breakdown of electrical insulator properties. Later, his work led to the Bell Labs implementation of the effect in form of an electronic device, the Zener diode.

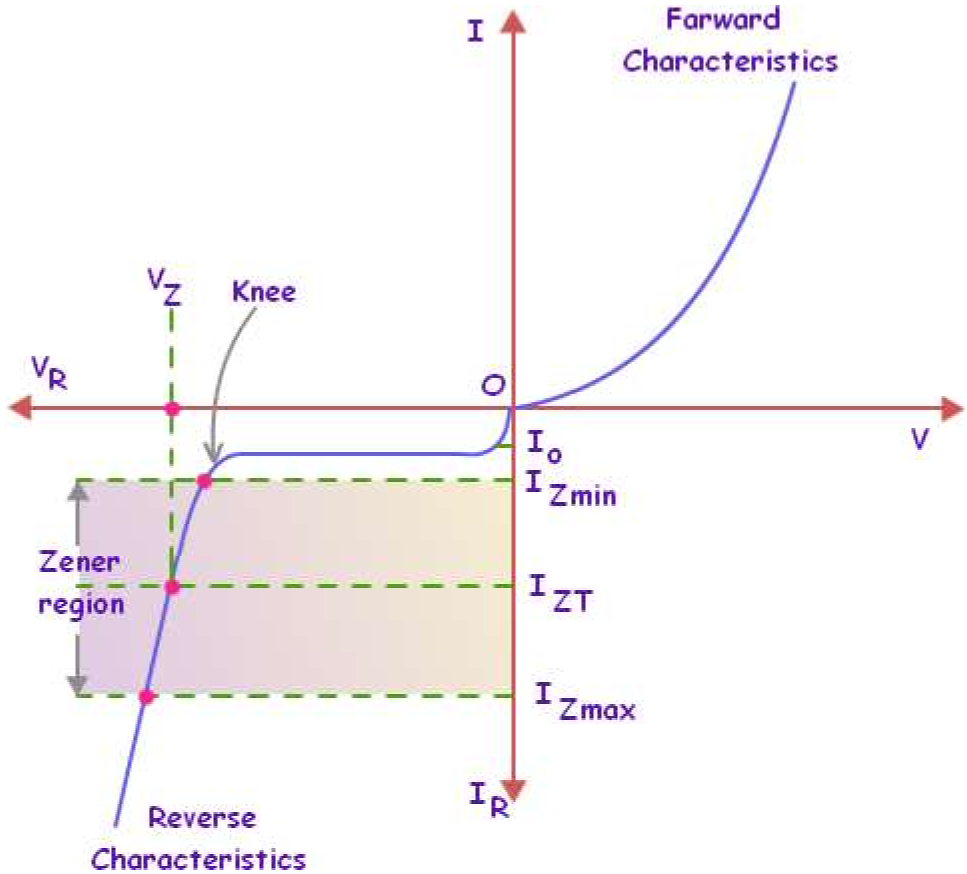


Fig. 3.8 IV Characteristics of a Zener diode

3.4 Bipolar Junction Transistor

A Bipolar Junction Transistor (also known as a BJT or BJT Transistor) is a three-terminal semiconductor device consisting of two p-n junctions which are able to amplify or magnify a signal. It is a current controlled device. The three terminals of the BJT are the base, the collector and the emitter. A BJT is a type of transistor that uses both electrons and holes as charge carriers.

If a signal of small amplitude is applied to the base is available in the amplified form at the collector of the transistor. This is the amplification provided by the BJT. Note that it does require an external source of DC power supply to carry out the amplification process.

There are two types of bipolar junction transistors – NPN transistors and PNP transistors. A diagram of these two types of bipolar junction transistors is given below (Figure 3.9):

Example 3.3

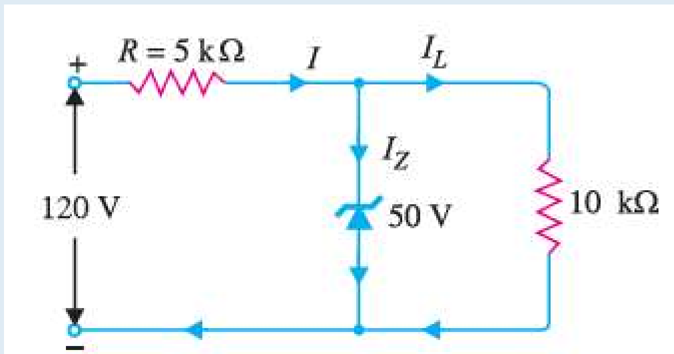
For the circuit shown in Figure

Find:

(i) the output voltage

(ii) the voltage dop across series resistance

(iii) the current through Zener diode.

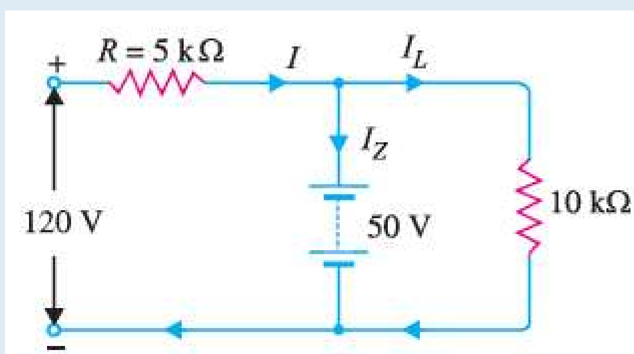


Solution:

If you remove the Zener diode in Figure, the voltage V across the open-circuit is given by:

$$V = \frac{R_L E_i}{R + R_L} = \frac{10 \times 120}{5 + 10} = 80 \text{ V}$$

Since voltage across Zener diode is greater than V_z (= 50 V), the Zener is in the "on" state. It can,therefore, be represented by a battery of 50 V as shown in Figure.



(i) Referring to Figure (ii)

$$\text{Load current, } I_L = V_Z/R_L = 50 \text{ V}/10 \text{ k}\Omega = 5 \text{ mA}$$

$$\text{Current through } R, I = \frac{70 \text{ V}}{5 \text{ k}\Omega} = 14 \text{ mA}$$

$$
\begin{aligned}
\text{Voltage across } 5\text{ k}\Omega &= 120 - 50 = 70\text{ V} \\
\text{Current through } 5\text{ k}\Omega,\ I &= \frac{70\text{ V}}{5\text{ k}\Omega} = 14\text{ mA} \\
\text{Load current, } I_L &= \frac{50\text{ V}}{10\text{ k}\Omega} = 5\text{ mA} \\
\text{Applying Kirchhoff's first law, } I &= I_L + I_Z \\
\therefore \quad \text{Zener current, } I_Z &= I - I_L = 14 - 5 = \mathbf{9\text{ mA}}
\end{aligned}
$$

Minimum Zener current:

The Zener will conduct minimum current when the input voltage isminimum i.e. 80 V. Under such conditions, we have,

$$
\begin{aligned}
\text{Voltage across } 5\text{ k}\Omega &= 80 - 50 = 30\text{ V} \\
\text{Current through } 5\text{ k}\Omega,\ I &= \frac{30\text{ V}}{5\text{ k}\Omega} = 6\text{ mA} \\
\text{Load current, } I_L &= 5\text{ mA} \\
\therefore \quad \text{Zener current, } I_Z &= I - I_L = 6 - 5 = \mathbf{1\text{ mA}}
\end{aligned}
$$

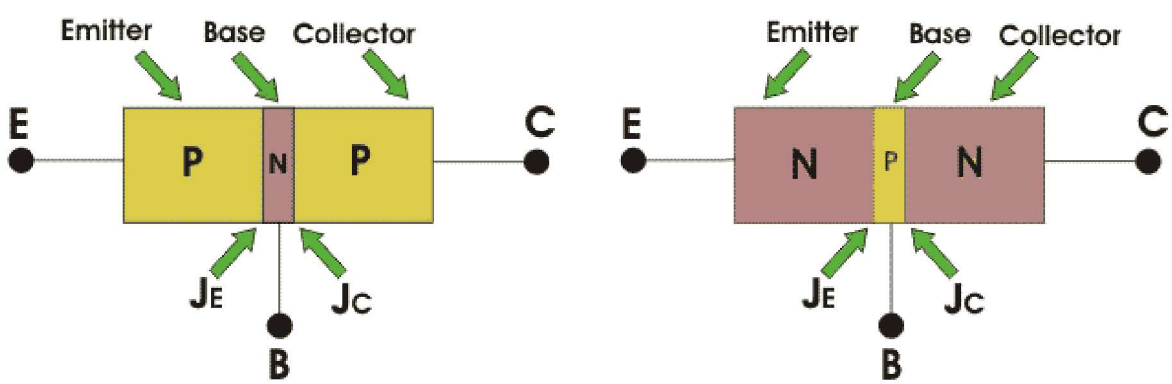


Fig. 3.9 IV Construction of BJT

From the above figure, we can see that every BJT has three parts named emitter, base and collector. J_E and J_C represent the junction of emitter and junction of collector respectively. Now initially it is sufficient for us to know that emitter-based junction is forward biased and collector-base junctions are reverse biased.

3.4.1 Applications of BJT

One common application of a BJT is to drive an LED. An LED driver is shown in Figure 3.10. The driver shown in this figure is used to couple a low current part of the circuit to a relatively high current device (the LED). When the output from the low current circuit is low (0 V), the transistor is in cut-off and the LED is off. When the output from the low current circuit goes high (+3.3 V), the transistor is driven into saturation and the LED lights. The driver is used because the low-current part of the circuit may not have the current capability to supply the 20 mA (typical) required to light the LED to full brightness.

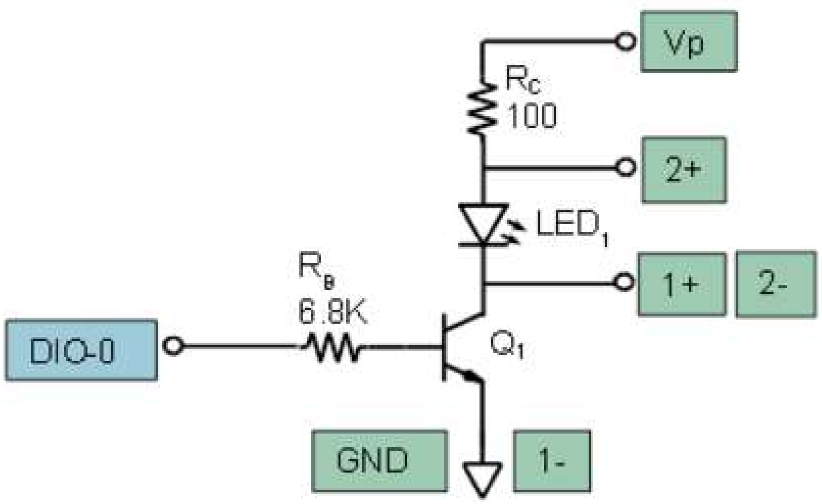


Fig. 3.10 BJT as a Switch

Activity

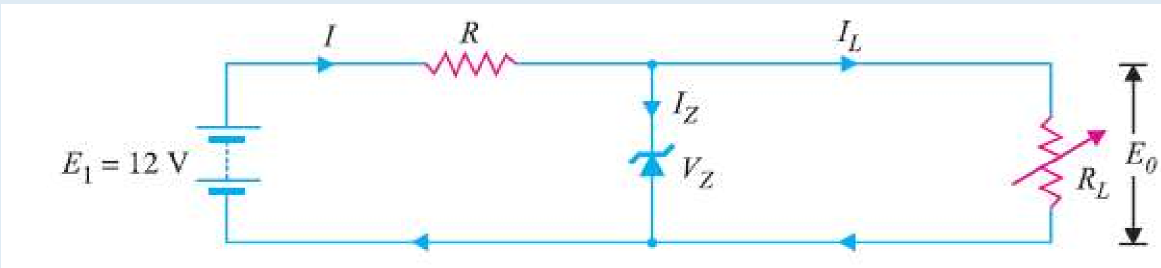
Build the LED switch circuit shown in figure 3.10 on your solder-less breadboard. R_C serves to limit the current that flows in the LED from the +5 V power supply (Vp). The switch is controlled by digital signal DO. Scope channel 1 will display the voltage across the switch transistor Q_1 (V_{CE}) and scope channel 2 will display the voltage across the LED.

Teacher Notes:

- Conduct this activity in Lab. Oscilloscope must be used to plot voltages.

Example 3.4

A 7.2 V Zener is used in the circuit shown in Fig. 3 and the load current is to vary from 12 to 100 mA. Find the value of series resistance R to maintain a voltage of 7.2 V across the load. The input voltage is constant at 12V and the minimum Zener current is 10mA.



Solution:

$$E_i = 12\text{ V}; \quad V_Z = 7.2\text{ V}$$

$$R = \frac{E_i - E_0}{I_Z + I_L}$$

The voltage across R is to remain constant at 12 − 7.2 = 4.8 V as the load current changes from 12 to 100 mA. The minimum Zener current will occur when the load current is maximum.

$$\therefore \quad R = \frac{E_i - E_0}{(I_Z)_{min} + (I_L)_{max}} = \frac{12\text{ V} - 7.2\text{ V}}{(10 + 100)\text{ mA}} = \frac{4.8\text{ V}}{110\text{ mA}} = 43.5\ \Omega$$

If R = 43.5 Ω is inserted in the circuit, the output voltage will remain constant over the regulating range. As the load current IL decreases, the Zener current IZ will increase to such a value that IZ + IL = 110 mA.

Note that if load resistance is open-circuited, then IL = 0 and Zener current becomes

Key Points

- A p-n junction is a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.
- The p-side or the positive side of the semiconductor has an excess of holes and the n-side or the negative side has an excess of electrons.
- A diode only conducts current in one direction.
- A diode is forward biased if a positive terminal of a source is connected to the p-type side and the negative terminal of the source is connected to the n-type side of the diode.
- A diode is reverse biased if we connect the negative terminal of the voltage source to the p-type side and the positive terminal of the voltage source to the n-type side of the diode.
- Rectifiers are used to convert AC signals to DC signals Flow of charges inside an electric circuit is called electric current.
- In a half-wave rectifier, one half of each AC input cycle is rectified.
- Full-wave rectifier circuits are used for producing an output voltage or output current which is purely DC.
- Zener diode is a heavily doped semiconductor diode which is designed to operate in reverse direction.
- Zener diode is commonly used for load regulation.

Exercise

Select the most appropriate option

1. ____ allows uni-directional flow of current.

   a. Resistance b. Capacitor c. Diode d. None of these

2. ____ is used for load regulation.

   a. Resistance b. Zener Diode c. Capacitor d. None of these

3. _____ can be used as a switch.

   a. BJT b. Capacitor c. BJT d. None of these

4. There are _____ p-n junctions in a BJT

   a. 1 b. 2 c. 3 d. Can be (a) and (c)

5. _______ is always used in reverse bias.

   a. Resistance b. Diode c. Zener diode d. BJT

Give short answer of the following

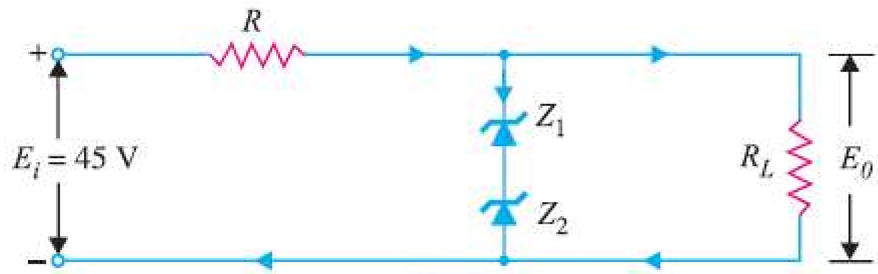
1. Define Diode.
2. What are the applications of diode?
3. Describe the formation of PN junction diode.
4. Define rectification.
5. Describe Half Wave rectifier.
6. Describe Full Wave rectifer.
7. Define Zener diode.

Answer the following question in detail

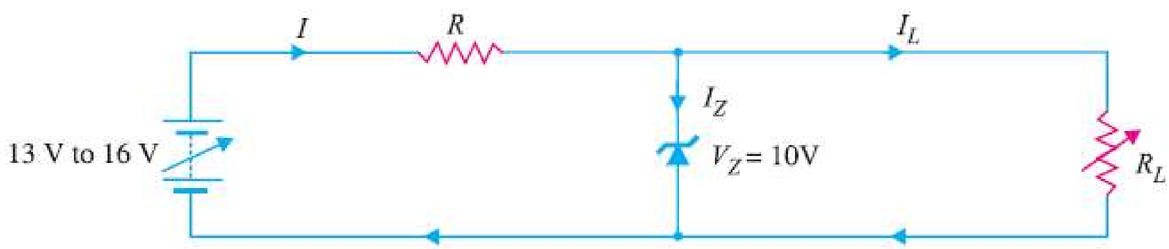
1. Describe the formation of zener diode.
2. Define load regulation and how zener diode can be used for load regulation.
3. What is the difference between Silicon and germinium based diodes.

Solve the Following

1. The circuit of Figure uses two Zener diodes, each rated at 15 V, 200 mA. If the circuit is connected to a 45-volt unregulated supply, determine :(i) The regulated output voltage (ii) The value of series resistance R.



2. A 10-V Zener diode is used to regulate the voltage across a variable load resistor. The input voltage varies between 13 V and 16 V and the load current varies between 10 mA and 85 mA. The minimum Zener current is 15 mA. Calculate the value of series resistance R.



3. The Zener diode shown in Figure has VZ = 18 V. The voltage across the load stays at 18 V as long as IZ is maintained between 200 mA and 2 A. Find the value of series resistance R so that E0 remains 18 V while input voltage Ei is free to vary between 22 V to 28V.